Hyperpigmentation in Various Conditions

Pre-existing pigmented lesions such as **nevi** and **ephelides** become darker during pregnancy. Also, recent scars often darken. In normally pigmented areas such as the **nipples**, **areolae**, and **genitalia**, the pigmentation becomes more intense. The **linea alba**, the median line on the anterior abdominal wall, often becomes hyperpigmented during pregnancy and is then called **linea nigra**. In a small proportion of pregnant women, hyperpigmentation occurs in the **axillae** or the **inner upper thighs**. **Melasma**, or "mask of pregnancy" (see the section Melasma), occurs in more than 50 percent of pregnant women (see Chap. 107).

Acanthosis Nigricans

Acanthosis nigricans is associated with hyperpigmentation and is discussed in Chaps. 152 and 154.

Diabetes

(See Chap. 153) A higher **vitiligo** prevalence has been reported in patients with **diabetes type I and II** compared to the general population. **Diabetic dermopathy** is characterized by asymptomatic, irregular, light brown, depressed patches on the anterior lower legs, but the pathogenesis is not known.

Nutritional Conditions

(See Chap. 130) Pigmentary changes can be secondary to nutritional conditions such as **kwashiorkor**, **vitamin B_{12} deficiency**, **folic acid deficiency**, or **pellagra**. The skin changes are reversible upon correction of the nutritional deficiency.

Metabolic Conditions

Porphyria Cutanea Tarda

Porphyria cutanea tarda is a metabolic disorder associated with diffuse brown hypermelanosis, accentuated in sun-exposed areas. In females, a melasma-like facial hyperpigmentation can be observed (see Chap. 132).

Hemochromatosis

Hereditary hemochromatosis is an autosomal recessive disorder associated with increased intestinal absorption of iron and deposition of excessive iron in the liver, pancreas, and other organs, including the skin. In the past, hemochromatosis was typically diagnosed at an advanced stage by the classic triad of **hyperpigmentation**, **diabetes mellitus** ("bronze diabetes"), and **hepatic cirrhosis**. Skin darkening was present in 70% of patients, due to two mechanisms:
- **Hemosiderin deposition**, causing a diffuse slate-gray discoloration
- **Increased melanin production**, causing a bronze hue

Because hemochromatosis is now usually diagnosed earlier, hyperpigmentation is less frequently observed. The pigmentation is typically generalized but may be more pronounced in sun-exposed areas, genitalia, and scars. Skin bronzing is reversible once phlebotomy therapy is initiated.

Tumoral Conditions

Mast Cell Disorders

Mast cell disorders can lead to hyperpigmentation and are discussed in Chap. 150.

Melanoma

(See Chap. 124) **Diffuse generalized melanosis** associated with advanced metastatic melanoma is a rare but well-documented event. It is characterized by a slate bluish-gray to brown discoloration of the skin. Histology reveals melanin particles and melanin-containing histiocytes and dendritic cells in the dermis and subcutaneous fat. No melanoma cells are detected in the skin, and there is no increase in epidermal melanin or melanocyte number. Circulating melanosomes have been detected in the blood, supporting the hypothesis that diffuse melanosis may result from tumor lysis with release of melanosomes into the circulation and subsequent deposition in the skin.

Physical Agents

Ultraviolet Radiation

The major acute effects of UV radiation on normal human skin are **sunburn**, followed by **tanning** (see Chap. 89).

Ionizing Radiation

(See Chap. 95) Exposure of the skin to ionizing radiation—such as in nuclear accidents (e.g., Chernobyl) or after local fractionated radiotherapy—can lead to **cutaneous radiation syndrome**, characterized by fibrosis, keratosis, telangiectasias, and sharply demarcated lentiginous hyperpigmentation resembling UV-induced lentigines.